Run report: matched-rho ecbench session `rho-01`

Study: `research/pdp_split_dreg_20261011` (PROTOCOL.md, "Matched rho"). **Descriptive only:** this page records what ran and what was measured; interpretation is in the study README.

What ran

- **Spec** `SPEC-rho.json`, id `ECS1h10c3ed7d3628`; **session** `ECBS1he23809badf5b`, directory `sessions/rho-01/`.
- **Command:** `ecbench run --wait --spec SPEC-rho.json --out sessions/rho-01 --cpus auto`.
- **Commit** `51bb1dc5e1fea5a7b53a2c6c8b02f51f75e77233` (clean); **binary** SHA-256 `909efb98…3d5740`.
- **Host:** Intel Xeon @ 2.10 GHz, 4 logical CPUs, 1 NUMA node, virtualised Linux 6.18 (Claude Code cloud container); runner CPUs 0–2. Started 2026-10-11 21:54:49 UTC, finished 12.4 s later.
- **Arms:** `rho-strong` = `rho.signed_frobenius_strong` (lanes 32, dp_bits 4, step_cap_factor 2000; reference); `rho-frob` = `rho.signed_frobenius` (cap_multiple 64; baseline).
- **Workloads:** Koblitz `icv1-f2m11-t67-05f5aa36`, `icv1-f2m13-t181-515ee569`, `icv1-f2m17-tm101-00378d4e`, `icv1-f2m19-tm797-9c54981b`; 8 public one-target workloads each (target seed 202610111101); 5 measured rounds + 1 warm-up; alternate order, seed 202610111102.

Completion and isolation

384 executions planned, 384 recorded, all `verified` (status counts `{"verified": 384}`). Isolation earned: L2 on most runs, L1 on 3–11 per (arm, curve) cell (blockers to L3: CPU not isolated, no performance governor). Wall time is not used by this study.

Audit: `ecbench verify --replay-all` → `audit OK: 384 records, 384 verified, 320 replays reproduced, 0 problems`; receipt `audits/rho-01.audit.json`, SHA-256 `0b56a9b730d9d14b726392d15500ffd22093413f048a72c4d1171099c33dcc0c`.

Measured values (ecbench table, unit S = charged group-addition equivalents / √r; all lower bounds by ecbench's rule, canonicalisations uncharged)

arm	curve	$\log_2$ r	mean S	95 % interval	S / floor
rho-strong	f2m11	9.95	42.024	21.121–77.100	157.3
rho-strong	f2m13	10.97	16.650	14.037–23.827	67.7
rho-strong	f2m17	16.00	4.054	3.913–4.201	18.9
rho-strong	f2m19	18.00	2.378	2.289–2.460	11.7
rho-frob	f2m11	9.95	2.448	2.366–2.575	9.16
rho-frob	f2m13	10.97	2.095	1.752–2.688	8.52
rho-frob	f2m17	16.00	1.093	0.914–1.330	5.09
rho-frob	f2m19	18.00	0.577	0.515–0.661	2.84

n = 40 verified runs per row (8 workloads × 5 rounds). Source: `audits/rho-01.table.md`; per-workload pairs in the session's saved comparison.

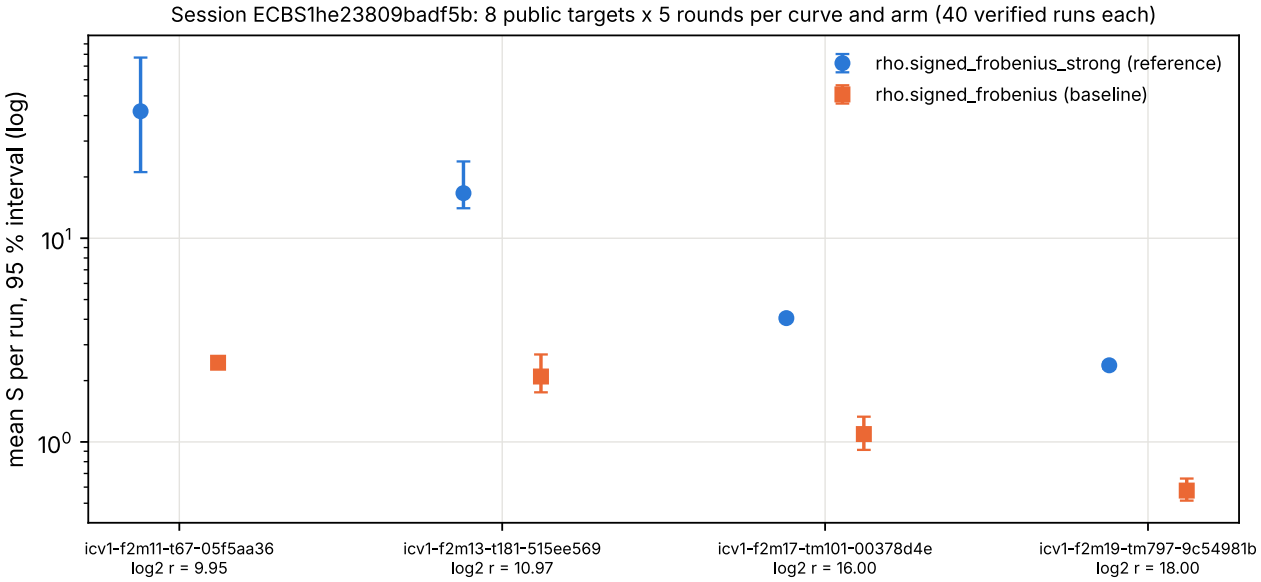


Figure 1: Mean S with ecbench's 95 % interval per arm and curve, session `ECBS1he23809badf5b` (recorded values; source `audits/rho-01.table.md`, layout `reports/plot_rho_session.py`).

Raw records

`sessions/rho-01/records.jsonl` (384 lines), `sessions/rho-01/session.json`, `spec.json`, `plan.json`, `host.json`; audit and table under `audits/`.